

## Element C: Presentation and Justification of Solution Design Requirements

At our independent boarding school we have an ongoing paper waste problem. Paper is being wasted because teachers and students sometimes print on only one side of the paper or make mistakes when using the printer or photocopier. This is a large-scale issue, over 200,000 pages have already been printed or photocopied this school year alone, and we still have three months until the end of the school year (Mr. B, personal communication, March 31, 2026). We need to find a solution to reuse the printed and photocopied paper to prevent it from being wasted and going into a landfill.

These are our prioritized design requirements to meet our stakeholders' needs:

- Safe to use - no sharp objects or edges - safe to touch (high priority)
- Durable (high priority)
- Affordable - less than \$2 per solution (high priority)
- Uses at least 90% of scrap paper (high priority)
- Uses 20 pages or more (medium)
- At least one side has to be blank (medium priority)
- 8.5 x 11 inches or smaller (low priority)

These will allow us to get to our solution and allow us to meet our stakeholders' needs. This will give our teachers a way to jot down notes and keep information all in one place.

Our product should be safe; it will not hurt students or teachers; that is very important because you cannot provide something that can hurt your stakeholders. We need to make sure the process is easy and affordable, and that we can make the

product. Making sure that most of the paper gets used is a high priority because if you do not actually help solve the problem, you have done nothing.

Since our solution is focused on safety to ensure that students or teachers cannot hurt themselves, we are also making sure to focus on affordability, as that is a key part of our solution. We also want to focus on using 90% of our scrap paper, with one side of each sheet blank. And because it is affordable and cost-effective, using inexpensive materials.

We communicated with our primary stakeholder, Mr. K, who is an engineering teacher at our independent school via email and agreed to most of the design requirements. Mr. K said the notebooks need to be uniform, all the same length, and at least 20 sheets. He told us that the binding of the solution that we originally put as a low priority should be put as a high priority.